

DS 301: Mathematical Foundations of Data Science

Course Code	DS 301
Course Name	Mathematical Foundations of Data Science
L-T-P-C	3-1-0-4
Prerequisites	IC110-Engineering Mathematics, IC111-Linear Algebra
Intended for	Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines
Approval	29th BoA (Mar 2019); 20th Senate (Apr 2019), in the B.Tech Data Science and Engineering curriculum

Course Contents

- **Module 1:** Definition of metric spaces, Examples, Open sets, Closed sets, Dense sets, Compact sets, Connectedness, Closure and interior of the sets, Metric subspace. (10 Lectures)
- **Module 2:** Cauchy sequences, Convergent sequences, Complete metric space, , Continuous functions, Continuity of composite functions, Continuity and inverse image of open and closed sets. (10 Lectures)
- **Module 3:** Normed linear spaces, Linear subspaces of normed linear spaces, Banach spaces, Riesz lemma, Continuity of linear maps, Bounded linear maps, Norm equivalence. (10 Lectures)
- **Module 4:** Hilbert spaces, Cauchy -Schwarz inequality, Parallelogram law, Orthogonality, Pythagorean Theorem, Orthogonal projection, orthogonal complement and projection theorem, Orthonormal sets, Orthonormal basis, Gram-Schmidt process, Examples of orthonormal basis. (12 Lectures)

Text Books

1. Apostol, T., **Mathematical Analysis**, 2nd Edition, Narosa Publishers, 2002.
2. Limaye, B. V., **Functional Analysis**, 2nd Edition, New age international Publishers, 2009.
3. Dan Simovici, **Mathematical Analysis for Machine Learning and Data Mining**, World Scientific, 2018
4. Rudin, W., **Principles of Mathematical Analysis**, 3rd Edition, McGraw-Hill, 2013.

References

1. Stein, E. M. and Shakarchi, M., **Real Analysis**, Princeton Lectures.
2. Tao, T, **Analysis I and II**, Trim, Hindustan book agency.
3. Kreyszig, E., **Introductory Functional Analysis with Applications**, Reprint, 2017
4. Naylor, A. C. and Sell, G. R., **Linear Operator Theory in Engineering and Science**.